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The notion of fractals in tumour angiogenic sprout initiation model based on cellular automata

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ABSTRACT

Tumour growth is considered chaotic, poorly controlled growth as its cells and vasculatures are irregular in shape. The vascular system of tumour is a complex network and its architectural complexity and geometrical characteristics cannot be defined with Euclidian geometry as it is only capable of expressing regular or smooth objects whereas, fractal geometry describes an object with irregularities and it is therefore well suited to quantify those morphological characteristics. In this paper, we develop a cellular automata (CA) based discrete model that tries to mimic all the features and make useful predictions of tumour angiogenesis with computer-coded rules. From the interactions of several species in 2-D tissue space, we study the effects of these various biological factors on the formation of the capillary sprouts. The model also captures anastomoses and branching phenomena in capillary blood vessels. It is observed that chemotaxis and randomness are the essential components for the capillary sprout motility whereas, haptotaxis works as an add-on feature. The model is also validated with partial sensitivity analysis. The fractal properties of the growing capillary sprouts in the model are investigated. The characterization of these fractal objects (capillary sprouts) is provided by the box-counting method. The model shows, the morphological change in the early stages of the vascularization process where the capillary sprouts are formed from the surrounding blood vessels. The model successfully simulates the morphological aspects of capillary vessel growth during tumour angiogenesis. The estimated fractal dimension is very close to the previously measured range. This is established our hypothesis about the fractal nature of vascular network pattern and our CA-based discrete model for capillary sprout initiation during tumour angiogenesis.

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1. Introduction

Angiogenesis refers to the formation of new blood vessels from the pre-existing vessels. It is a fundamental process that can be seen in some of the physiological processes like reproduction, embryonic development, wound healing, and in some of the pathological processes like, vascularization of ischemic tissues [1], rheumatoid arthritis [2], diabetic retinopathy [3], chronic inflammation, and tumour growth, its progression [4] and metastasis [5], etc.

At the early stage, a tumour nodule does not have any direct blood vessels support; so it has to depend on the host's vasculature for the nutrients, oxygen, and other growth elements. The metabolic demand of a tumour rises with its volume [6], however, it takes nutrients and oxygen proportionally to its surface area. As a result, tumour cells near the center become hypoxic due to a lack of oxygen and nutrients. At this condition, local pH balance is disturbed, and oxygen and nutrient concentrations in the surrounding microenvironment are decreased sharply [7]. In this situation, the growth of a tumour would be stagnant (1–2 mm of diameter). Without direct access of blood vessels, the hypoxic tumour cells gradually become necrotic. Hence, for continuing its growth, the tumour shifts to an angiogenic phenotype and attracts surrounding blood vessels toward it.

Hypoxia Inducible Factors (HIFs) are the main controller of the hypoxic response. HIF protein family belongs to Helix–Loop–Helix PER–ARNT–SIM (b–HLH–PAS) family which consists of $\alpha\beta$–heterodimers. HIF protein family includes HIF–1α, HIF–2α, HIF–3α, and HIF–1β. Activation of HIF directly affects tumour growth as it is one of the key regulators of blood vessels' development, maturation, and remodelling. Only HIF–1α controls oxygen homeostasis in many cell types [8], and it is a very influencing promoter for tumour growth as it controls the activity of endothelial cells (ECs). Researchers have found a correlation among HIF–

Abbreviations: CA, Cellular automata; HIF, Hypoxia inducible factor; EC, Endothelial cell; VEGF, Vascular endothelial growth factor; FGF, Fibroblast growth factor; ANG, Angiopoietin; PLGF, Placental growth factor; VEGFR, Vascular endothelial growth factor receptor; MMP, Matrix metalloproteinases; ECM, Extracellular matrix.

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1α and HIF–2α with the activation of many pro-angiogenic factors like vascular endothelial growth factors (VEGFs), acidic- and basic fibroblast growth factors (a–, b–FGFs), placental growth factor (PLGF), angiopoietin–1, 2, and 4 (ANG–1, 2 and 4), and etc. [9]. These influences destroy the balance between the pro– and anti-angiogenic factors. Among these pro-angiogenic factors VEGF is highly expressed in most of the mammalian tumour in ECs [10].

VEGF consists of six members in the family which includes VEGF–A to E, and PLGF that communicate with tyrosine kinases, vascular endothelial growth factor receptor-1 (VEGFR–1), and VEGFR–2 to alter angiogenesis. Scientists have confirmed that the interplay between VEGF–A and VEGFR–2 are the major contributors to sprouting angiogenesis [11]. VEGF–A is a highly specific mitogen for vascular ECs that leads to the stimulation of the entire process required for neovascularization. It is found that the level of VEGF is very high in tumours. Tumour cells secret VEGF from their surrounding environment. It is used for the nourishment of the new blood vessels. VEGF also up-regulates plasminogen activators (PAs) [12], and matrix metalloproteinases (MMPs) [13] during tumour angiogenesis. On the other hand, FGFs are heparin-binding factors. FGFs and their receptors (FGFRs) control growth and differentiation of healthy cells (non-cancerous). FGF family consists of 23 members. Among them, a–FGF (acidic) (FGF–1), and b–FGF (basic) (FGF–2) are identified as the pro-angiogenic factors which influence EC proliferation and migration [14]. The up-regulation of FGF and FGFRs is found in tumour cells. It is a very important factor for the angiogenesis cascade, though it does not have signaled sequences that trigger secretion. FGF is secreted in the extracellular matrix (ECM); after that angiogenesis starts. In angiogenesis, FGF works with VEGF in cooperated manner. So, the interaction between FGF and VEGF is also an important issue in tumour angiogenesis [15].

The angiopoietin family consists of four members (identified till now), among which ANG–1, ANG–2, and ANG–4 are involved in angiogenesis. ANG–1, 2, and 4 bind with endothelial tyrosine kinase receptor: Tie–2. The Angiopoietin family restricts several ligands to bind with the same receptors. While ANG–1 activates Tie–2 signal, ANG–2 opposes its activation. The functions of ANG–1 are EC migration, adhesion, and pericyte recruitment; on the other hand, ANG–2 destabilizes the vessels [16]. The PDGF family consists of five members: PDGF–A to D, and PDGF–AB. These proteins activate the downstream signals through binding PDGFR tyrosine kinases receptors and activating molecules like VEGF [17,18]. PDGFs are mainly involved in the growth and progression of various kinds of cells. Scientists found a strong correlation between the up-regulation of PDGF, with FGF and VEGF [19,20]. PDGFs are involved in the recruitment of pericytes to the blood vessels, enhancement of secretion of other pro-angiogenic factors, endothelial proliferation and motility, sprouting, and tube formation [21].

These pro-factors diffuse through the ECM and degrade the basement membrane of nearby blood vessels. Initially, the TAFs disrupt the corresponding EC receptors which have a direct influence on the structure of capillary networks. Finger-like capillary sprouts are created from the disrupted areas of the surrounding blood vessels [22]. These capillary sprouts are grown by the proliferation of recruited ECs from the parent vessels. The TAFs generate a chemical gradient between the tumour and the growing capillary vessels; so that, the capillary sprouts migrate toward the tumour.

ECs of the capillary sprouts synthesize and secrete cellular fibronectin. Fibronectin does not diffuse in the environment but increases adhesiveness between the capillary sprouts and the ECM [23]. On the other hand, VEGF upregulates MMPs expression in the ECs which remodel the ECM [13]. After covering a certain distance towards the tumour, capillary sprouts may often incline toward each other and fuse together to form loops (anastomoses). These loops are the milestone of blood circulation within the capillary vessel network. From these loops, new sprouts may generate and the same sequence of events is repeated for further extension of the network. New capillary sprouts can also be generated from a sprout tip (branching). These processes are occurred continuously and formed a complex vascular network.

In physiological angiogenesis, pro-and anti-angiogenic factors are in balance that regulates the growth of the newly developed vessels, whereas, this balance is lost in tumour angiogenesis and the pro-factors become more powerful. As a result, newly developed vessels grow unboundedly and the vascular network becomes unorganized, often leaky, chaotic in nature, and hemorrhagic due to upregulation of growth factors. Blood flow within these networks is often slow or oscillating which makes it dysfunctional [24].

Angiogenesis is an essential phase for the tumour cell to metastasize as there exists a strong correlation among tumour microvessels density (MVD) with the metastatic potential ability and the poor survival rates [25,26,27]. In tumour angiogenesis, newly developed capillaries are growing unboundedly. These vessels often carry tumour cells in their wall [28]. The leaky vessels enable hematogenous spread of cancer cells either singly or collectively [29]. After detaching from the primary tumour, cancer cells may enter into the leaky vessels and spread all over the body to metastasize the remote organs. In cancer, almost ninety percent of patients have died of metastasis.

Computational oncology encompasses any form of computer-based modelling relating to tumour biology and cancer therapy. The aim is to build computer models that simulate biological processes and use these models to make useful predictions. Researchers have developed various models to explain tumour growth, as well as its progression from different perspectives as most of the processes involved in tumour growth, are not clearly understood. The work of Anderson and Chaplain [30] is one of the most cited article in tumour angiogenesis modelling. In this study, they have presented both continuum as well as discrete models for capillary sprout formation during tumour angiogenesis. Both of these models consider the interactions between ECs of blood vessels and the ECM via fibronectin. They have concluded that chemotaxis and haptotaxis are the necessary factors for the development of capillary network at a large scale and the chemotactic response should be strong enough for the initial outgrowth of the capillary network.

Sleeman and Wallis [31] have proposed a reinforced random walk-based model that incorporated chemotaxis and haptotaxis movements of the ECs. They observed anastomoses without proliferation of the EC and developed a realistic blood vessel network. Plank and Sleeman [32] extended the prior research of Sleeman and Wallis [31] by including the concept of anti-angiogenic strategy. They have examined the ability of angiostatin to restrict capillary growth. They observed that, proteolysis and proliferation are crucial processes for tumour vascularization whereas, angiostatin is proficient to restrict capillary growth. Milde et al., [33] have developed a hybrid model of sprouting angiogenesis. The model combines continuum approximation of several TAFs, fibronectin, and the EC density with discrete particle-based sprout tip cells. The capillary vessels are represented by the continuous grid-independent particle mesh interpolation method. Also, filopodia are modelled explicitly to sense the migration and branching direction. The model explains that the structure and the density of the ECM have direct impact on the morphology, EC migration, expansion speed, the number of capillary branches, and network formation in the newly developed vessels.

Bentley et al., [34] have developed an agent-based model (ABM) for sprout cell stimulation to investigate the tip-stalk cell pattern morphology. The model assumed VEGF-A gradient and filopodia extension are the important parameters for tip-stalk cell modelling.

They showed that upregulation of VEGF-A produces defects in capillary vessels, increases tip cell selection rate, and oscillates VEGFR-2 along a vessel. Bentley et al., [35] have proposed another ABM for tip cell motility. In this study, they have explored the tripping point between normal and abnormal sprouting due to high VEGF concentration. They also showed that tip-stalk cell can interchange their roles during fusion. The article has concluded to use migration inhibitors agents for vessel normalization in cancer treatment. Wcisło et al., [36] have developed a 3-D multi-scale cellular automata (CA) model based on particle dynamics for tumour angiogenesis. The tissue and fragments of the vascular network are formed with particles and interact with the surrounding neighbours via mechanical resistance of the cell wall. The model can be able to explain the realistic 3-D dynamics of the entire tumour microenvironment consist of normal and cancerous cells, blood vessels, and blood flow.

Perfahl et al., [37] have investigated a 3-D multi-scale model of vascular tumour growth to find out the roles of the domain size, aspect ratio, and initial vascular network on tumour growth. They found that the smaller part of a large domain (subdomain) never represents the whole large domain. They have also shown that the extrapolation of results from the subdomain to a large domain is only possible if the size of the subdomain is sufficiently large enough and initialized with a complex vascular network. Daub and Merks [38] have developed a cellular Potts model for tumour angiogenesis that includes a set of biologically motivated cell behavioural rules for developing sprouts and branching in the vascular network. Lyu et al., [39] have shown that the capillary network has grown gradually as the tumour grows during the vascularization process. They have also shown that tumour is divided into necrotic, semi-necrotic, and well-vascularized regions.

Vilanova et al., [40] have focused on vessel remodelling during angiogenesis due to the absence of blood flow. They also provided the regression mechanism based on the tumour induced capillary vessel network. Salimi Sartakhti et al., [41] have developed a model based on evolutionary game theory to explore the roles of non-cooperative tumour cells present in the tumour during angiogenesis. In this context, interactions and public goods diffusion are described through a graph to illustrate the spatial configuration of tumour during angiogenesis. Hendrata and Sudiono [42] have developed a hybrid multi-scale model for investigating angiogenesis and tumour growth under mesenchymal stem cell (MSC) mediated therapy. The model has shown the effectiveness of MSCs in EC apoptosis that depends on MSC density. They claimed that MSC therapy has shown its full effect within several days. Phillips et al., [43] have presented a hybrid mathematical model for vascular changes via an ABM. The model has shown the physical interactions between a tumour and local blood vessels. It also successfully captured collapsing and breaking of vessels caused by tumour-EC interactions.

All of these models capture tumour-induced angiogenesis from different aspects but none of them have nurtured its fractal characteristics except Grizzi et al., [44]. Cancer is considered as chaotic, poorly controlled growth as its cells and vasculatures are irregular in shape. The vascular system of tumour is very complex network and its architectural complexity and geometrical characteristics cannot be defined with Euclidian geometry as it is only capable of expressing regular or smooth objects like lines, surfaces, planes, or spheres. Fractal is a structure that shows self-similarity. It may not be the same structure at all the scales but the same type of structure must appear in all the scales. Fractal geometry describes an object with irregularities [45] and it is therefore well suited to quantify those morphological characteristics. Pathologists have used this feature of fractal geometry qualitatively to describe malignancies; for example, their ragged border with host tissue or random pattern of growing capillary vessels, etc. Grizzi et al., [44] have used surface fractal dimension to describe 2-D geometrical complexity of tumour vasculature for studying vessel density and distribution during computer simulation. They have shown surface fractal dimension is dependent on the number of vessels and their distribution which indicates that quantitative evolution of the 2-D geometrical complexity is useful for the measurement of complex architecture as well as its development and growth.

CA is intensively used for modelling complex natural systems that contain of a large number of simple identical subparts with chaotic behavior [46]. In this paper, we develop a CA-based discrete model that tries to mimic all the features of tumour angiogenesis with computer-coded rules. We also investigate the fractal properties, the number of branching, and anastomoses of the growing capillary sprouts in the model. The characterization of a fractal set is provided by the box-counting method [47,48]. The model shows, the morphological change during the early stages of vascularization where the capillary sprouts are formed from the nearby blood vessels. The model contains a set of computer-coded discrete rules that define the spatial and temporal changes of capillary sprout networks and their surrounding chemical components.

The proposed CA model focuses on different factors which are involved in tumour angiogenesis, like, VEGF, FGF, PDGF, ANG, fibronectin, MMPs, and the ECs of the surrounding blood vessels. We develop a set of rules for these factors. The model also considers the chemotactic and haptotactic responses of the capillary sprouts within the domain of study. These rules update the 2-D grid of tissue simultaneously. Due to chemotactic and haptotactic forces, capillary sprouts move toward the tumour using EC proliferation. The movement of these sprouts is controlled by the tip cells. The model also includes capillary sprout branching and anastomoses. The outcome of the model justifies the morphological characteristics of the capillary vessel network and provides an opportunity to analyse the network for different quantitative measurements.

The model has successfully simulated the morphological aspects of capillary vessel growth during tumour angiogenesis. The fractal forms generated from the CA model lead to self-similar patterns. Here, we try to draw a relationship between the simulated forms and real capillary network patterns through the estimation of fractal dimension. The fractal dimensions are estimated for the two outcomes that are very close to the usual range [45]. This conclusion establishes our hypothesis about the fractal nature of vascular network patterns and our CA-based discrete model for sprouting angiogenesis.

The model is validated using sensitivity analysis. The purpose of the sensitivity analysis is to quantify the variability in the model due to the input parameters; *i.e.*, to measure robustness in the model by changing one or more input parameters. In this article, we quantify the morphological characteristics of the capillary vessel network by measuring *the number of anastomoses*, *number of branches*, *number of final capillary sprouts*, and *mean fractal dimension.*

This paper is organized as follows: Section 2 discusses about tumour vascularization and fractals; Section 3 illustrates the CA based discrete model of tumour angiogenesis; Section 4 describes the parameter estimation and computer simulation of the model; Section 5 estimates the fractal dimensions of the model outcomes; Section 6 discusses the validation part of the model; and Section 7 concludes this article.

2. Tumour vascularization and fractals

The evolution of tumour is a very complex process that includes several events across different scales and phases. Understanding the tumour growth dynamics is a great challenge. Tumour vascularization is one of the pathological processes by which tumour gets its vascular network from the host tissue. The vascular net-

work ultimately provides blood supply within the tumour and acts as an unlimited source of nutrients and glucose for further growth of the tumour. Due to the rapid growth of tumour, mechanical pressure within the tumour centre creases that may be the reason for blood vessels' occlusion and thereby restrict drug delivery to the tumour.

Scientists have identified tumour vascularization as a critical phase in tumour growth and a potential target for the therapeutic purpose [49]. The motivation behind this observation is that if a tumour is unable to build its vascular network, then it will not succeed to increase its size beyond a diameter of few millimeters. We have acquired a good knowledge of the process of angiogenesis at molecular level over the last two decades, but we fail to get the same for organ or tissue levels. The vascular network of tumour is completely diverse from vasculature in normal tissue as it is known to be more chaotic [45,50] due to loops, branches, shunts, variable intravascular distances, and large avascular areas [50]. Vasculature in normal tissue shows tree-like branching patterns or constant intravascular distances. However, no such feature exists in tumour vascular network. Understanding the differences behind these morphological variations may be a milestone for improving drug delivery to a tumour.

Tumour vasculature can be geometrically represented as a complex fractal network that irregularly branches with a systematic reduction in their length and diameter. The reasons behind the differences between tumour vasculature and the normal vascular network are still unknown although the same growth factors and inhibitors are involved in both cases. The tumour angiogenesis is identified as a critical phase in tumour growth. Hence, a consistent effort is required to identify its origin and the reasons for an abnormality that is crucial for treatment purposes [49]. Controlling the tumour vascularization process may regulate tumour growth. So, researchers have been targeting the growth of tumour vessels for the last 30 years. Moreover, identification of different anti-angiogenic substances makes this goal easier [49,51]. For understanding the ability of the vessels to deliver drugs or transport nutrients, it is very much required to know the pattern and the number of vessels creation in the vascular network.

Fractal objects can have the following four properties: (i) irregularity in shape, (ii) self-similarity in the structure, (iii) non-integer or fractal dimension, and (iv) scaling [44]. Self-similarity is a special feature of fractal objects that shows the repetition of the similar pattern at decreasing orders of magnitude [52]. Hence, the form of their components is similar to the whole. While geometrical self-similarity concern about the mathematical fractal objects where every small part is identical to the whole, statistical self-similarity has taken care of all complex systems including tumour vasculature. Scaling describes the measured properties depend on the scale on which they are measured.

Fractal analysis is a tool for measuring complex structures. It can simplify a complex structure like tumour vascular network into a set of numbers that describe its self-similar nature [53]. In general, vessel length, diameter, branching angle, etc. are used for describing traditional networks [54], whereas, the fractal analysis produces crucial information regarding the formative process of tumour vasculature [55]. In short, fractal analysis distinguishes among local, diffusion-limited, and compact growth processes of vessels.

3. Model formulation

In this section, we describe the ideas and the assumptions that help to build the CA model in a 2-D framework as the model in 3-D is computationally expensive. Though, the 3-D version of the model with comparatively small tissue space is discussed in the Appendix. We found that, outcome of both (2-D and 3-D models) are very similar in nature. In this article, we mainly focus on the morphological characteristics, such as, the trajectory of the sprout tips or the shape of the capillary sprouts (as the capillary sprouts are guided by the corresponding tip cells), number of anastomoses, number of branches, number of vessels reach to the tumour at end of simulation.

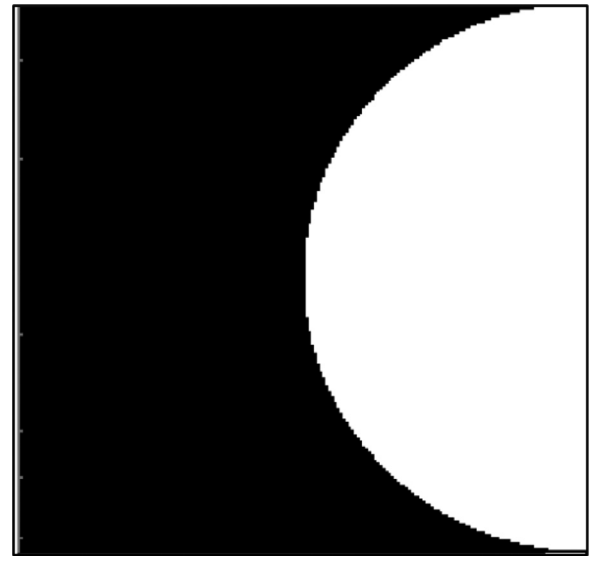

Fig. 1. Initial positions of tumour and existing blood vessel. On the right side of the figure, white half circle represents the tumour while on the opposite site the thin white line represents the existing blood vessel.

We consider a spatial domain $\mathbf{\Phi}$ that consists of the primary tumour and its surrounding microenvironment. In this spatial domain, we present VEGF, FGF, PDGF, ANG, fibronectin, and MMPs as the chemical concentrations which changes depending upon computer coded rules with time instances. Here, we ignore the tumour cell dynamics in the spatial domain. Here, we consider the tumour as a static entity as its growth will be stagnant after it becomes 1–2 mm in diameter [22,11,56,57]. We also consider EC of the existing blood vessels as a concentration which present at the opposite direction of the tumour. Therefore, in the domain $\mathbf{\Phi}$, existing blood vessel and the tumour both present along the Y-axis but in the opposite directions Fig. 1). The model considers that, initially tumour cell secrete VEGF [C_V], FGF [C_F], PDGF [C_P], and ANG [C_A] that travel all the way along through the X-axis and reach to the existing blood vessel at the opposite direction. Hence, the level of these concentrations near the tumour are very high and it gradually reduces as it progress toward the blood vessel. On the other hand, ECs secrete fibronectin [C_f] that travels exactly in the opposite direction of the TAFs. Therefore, the level of fibronectin near to the blood vessel is very high and it reduces gradually as it approaches to the tumour (Fig. 2(a), (b)). FGF, PDGF, and ANG produce the same kind of plot like Fig. 2(a), so these plots are not shown here. Hence, each and every chemical components are initialized properly using Eqs. (1)–((5). Eqs. (1) and (5) are taken from Anderson and Chaplain [30].

$$[C_V]_{i,j}^{t=0} = \sigma_1 \exp\left(\frac{-(1-x)^2}{\varepsilon_1}\right), (x,y) \in [0,1] \times [0,1] \tag{1}$$

$$[C_F]_{i,j}^{t=0} = \sigma_2 \exp\left(\frac{-(1-x)^2}{\varepsilon_2}\right), (x,y) \in [0,1] \times [0,1] \tag{2}$$

$$[C_P]_{i,j}^{t=0} = \sigma_3 \exp\left(\frac{-(1-x)^2}{\varepsilon_3}\right), (x,y) \in [0,1] \times [0,1] \tag{3}$$

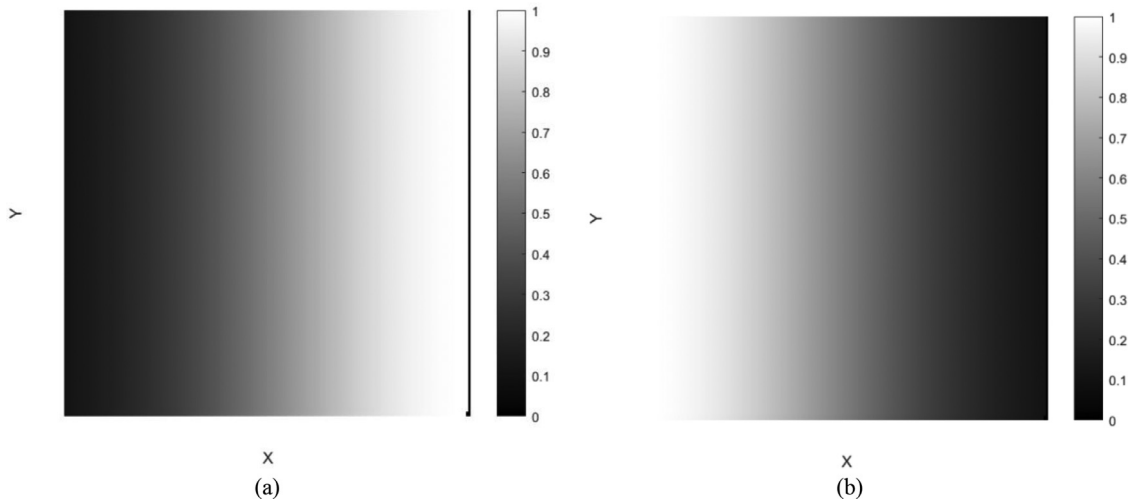

Fig. 2. Initial concentration levels of (a) VEGF and (b) fibronectin.

$$[C_A]_{i,j}^{t=0} = \sigma_4 \exp\left(\frac{-(1-x)^2}{\varepsilon_4}\right), (x, y) \in [0, 1] \times [0, 1] \tag{4}$$

$$[C_f]_{i,j}^{t=0} = \sigma_5 \exp\left(\frac{-x^2}{\varepsilon_5}\right), (x, y) \in [0, 1] \times [0, 1] \tag{5}$$

In Eqs. (1)–(5), σ_1, σ_2, σ_3, σ_4, σ_5, ε_1, ε_2, ε_3, ε_4, and ε_5 are the positive constants. The model gives the maximum weightage to VEGF among all the TAFs, as it is the most effective and important chemical concentration required for tumour angiogenesis [11]. It is also considered that, except fibronectin all the chemical concentration involved in angiogenesis are transported through diffusion. During angiogenesis, TAFs stimulate the ECs of the existing blood vessels. Finger like capillary sprouts are formed from the EC lines. Further, secreted TAFs up regulate the expression of MMPs (C_M) within the ECs of capillary vessels. The growing capillary sprouts secretes MMPs in the microenvironment that remodel fibronectin. By assuming all the above details we model the transportation process of the TAFs, fibronectin, MMPs, and the ECs with Eqs. (6)–(11).

$$[C_V]_{i,j}^{t+1} = [C_V]_{i,j}^{t} + \alpha_1.\text{avg}\left(\text{Mo}\left([C_V]_{i,j}^{t}\right)\right) - \alpha_2[C_V]_{i,j}^{t}[\text{Ec}]_{i,j}^{t} \tag{6}$$

$$[C_F]_{i,j}^{t+1} = [C_F]_{i,j}^{t} + \alpha_3.\text{avg}\left(\text{Mo}\left([C_F]_{i,j}^{t}\right)\right) - \alpha_4[C_F]_{i,j}^{t}[\text{Ec}]_{i,j}^{t} \tag{7}$$

$$[C_P]_{i,j}^{t+1} = [C_P]_{i,j}^{t} + \alpha_5.\text{avg}\left(\text{Mo}\left([C_P]_{i,j}^{t}\right)\right) - \alpha_6[C_P]_{i,j}^{t}[\text{Ec}]_{i,j}^{t} \tag{8}$$

$$[C_A]_{i,j}^{t+1} = [C_A]_{i,j}^{t} + \alpha_7.\text{avg}\left(\text{Mo}\left([C_A]_{i,j}^{t}\right)\right) - \alpha_8[C_A]_{i,j}^{t}[\text{Ec}]_{i,j}^{t} \tag{9}$$

$$[C_f]_{i,j}^{t+1} = [C_f]_{i,j}^{t} + \omega_1[\text{Ec}]_{i,j}^{t} - \omega_2[C_f]_{i,j}^{t}[\text{Ec}]_{i,j}^{t} \tag{10}$$

$$[C_M]_{i,j}^{t+1} = [C_M]_{i,j}^{t} + \gamma_1.\text{avg}\left(\text{Mo}\left([C_M]_{i,j}^{t}\right)\right) + \gamma_2.\psi(i, j)[\text{Ec}]_{i,j}^{t} - \gamma_3[C_M]_{i,j}^{t}[\text{Ec}]_{i,j}^{t} \tag{11}$$

In these Equations 6 – 11, α_1, α_2, α_3, α_4, α_5, α_6, α_7, α_8, ω_1, ω_2, γ_1, γ_2, and γ_3 are the positive parameters. In this model, we consider that, the level of all the chemical concentrations (except fibronectin) at the grid point (i, j) is affected by the levels of its diagonal as well as orthogonal grid points. In this context, we modify the level of concentrations at (i, j) by the average value of its Moore's neighbors. In the Eqs. (6)–(9) and (11), Mo($A_{i,j}$) refers to the Moore's neighbors elements of (i, j)th position in the 2-D grid A, *i.e.* $A_{i\text{-}1,j\text{-}1}$, $A_{i\text{-}1,j}$, $A_{i\text{-}1,j+1}$, $A_{i,j+1}$, $A_{i+1,j\text{-}1}$, $A_{i+1,j}$, $A_{i+1,j\text{-}1}$, and $A_{i,j\text{-}1}$ Fig. 3(a)). Moreover, it is considered that EC concentration is updated by its four orthogonal grid points only. Hence, for updating EC concentration, we are taking the average of its von Neumann neighborhood. Therefore, in ((12), vN($A_{i,j}$) refers to the von Neumann's neighbors of (i, j) in the 2-D grid A, *i.e.* $A_{i\text{-}1,j}$, $A_{i,j+1}$, $A_{i+1,j}$, and $A_{i,j\text{-}1}$ (Fig. 3(b)).

$$[\text{Ec}]_{i,j}^{t+1} = [\text{Ec}]_{i,j}^{t} + \lambda.\text{avg}\left(\text{vN}\left([\text{Ec}]_{i,j}^{t}\right)\right) .\left(\varphi_1[C_V]_{i,j}^{t} + \varphi_2[C_F]_{i,j}^{t} + \varphi_3[C_P]_{i,j}^{t} + \varphi_4[C_A]_{i,j}^{t}\right) \tag{12}$$

In (12), λ, φ_1, φ_2, φ_3, and φ_4 are the positive parameters. φ_1, φ_2, φ_3, and φ_4 are the weightage given for the respective TAFs considered here, such that, $\varphi_1 > \varphi_2 > \varphi_3 > \varphi_4$, and $\varphi_1 + \varphi_2 + \varphi_3 + \varphi_4 = 1$.

The movement of these capillary sprouts are controlled by their corresponding tip cells. The newly developed capillary sprouts (S) move towards the tumour due to chemotactic force (for higher concentration of TAFs) while fibronectin increases the adhesiveness between the sprouts and the ECM that generate a haptotactic force between them. Therefore, the movement of capillary sprouts are controlled by the chemotactic as well as the haptotactic forces. As the capillary sprouts approach towards the tumour, it can move either adjacent eight directions (Moore neighbors) or it may remain at the same position. In this context, randomness is also included in this model by incorporating a random variable. Considering all the mentioned features above, we produce Eq. (13) as,

$$S_{i,j}^{t+1} = \text{indexof}\left\{\max\left(g_{i,j}^{t}, g_{i-1,j}^{t}, g_{i+1,j}^{t}, g_{i,j-1}^{t}, g_{i,j+1}^{t}, g_{i-1,j-1}^{t}, g_{i-1,j+1}^{t}, g_{i+1,j-1}^{t}, g_{i+1,j+1}^{t}\right)\right\} \tag{13}$$

Where

$$g_{i,j}^{t} = \underbrace{w_1\left(\varphi_1[C_V]_{i,j}^{t} + \varphi_2[C_F]_{i,j}^{t} + \varphi_3[C_P]_{i,j}^{t} + \varphi_4[C_A]_{i,j}^{t}\right)}_{\text{chemotaxis}} + \underbrace{w_2[\text{Ec}]_{i,j}^{t}[C_f]_{i,j}^{t}}_{\text{haptotaxis}} + \underbrace{w_3 pd_{i,j}}_{\text{randomness}} \tag{14}$$

Eq. (13) indicates that, the sprout tip located at (i, j) will move at next time instance to the location that has the higher 'g' value.

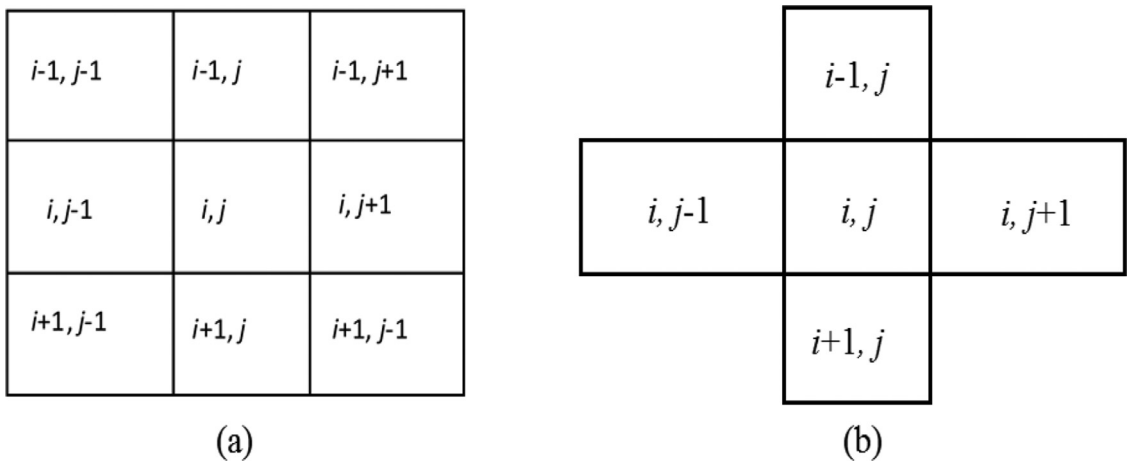

Fig. 3. (a) The Moore's neighborhood, and (b) von Neumann's neighborhood of a 2-D grid position (i, j).

If the Eq. (13) returns (i, j), then it indicates that the sprout tip will be in idle position otherwise the sprout moves to a neighboring (Moore neighborhood) specified position. In (14), w_1, w_2, and w_3 are the corresponding weightage given to chemotaxis, haptotaxis, and randomness respectively, where, $w_1 + w_2 + w_3 = 1$. p is a random variable ($p \in [0, 1]$with uniform distribution) in (14), which infuses randomness in the equation, and $d_{i,j}$ is reciprocal of the Euclidian distance between the tumour centre (1, 0.5) and the point (i, j).

If the moving capillary sprouts find the tip cells of two capillary vessels meet at the same position then they formed a close loop by fusing the tip cells and anastomoses happened with probability P_a. On the other hand, new vessels can be created from the existing sprout tip (branching) depending on some reasonable assumptions. Without loss of generality, it is considered that for branching, the concentration of ECs must be available in sufficient quantity. There should be available vacant spaces for newly developed branches. The last and most important condition for branching is that, the tip cell must be mature enough to split. Therefore, from these assumptions we use the following conditions for vessel branching,

(i) The concentration of EC must be greater than $[\mathrm{Ec}]_{th}$.
(ii) The *age* of the sprout tip should be greater than the threshold age_{th}.
(iii) There should be sufficient space available for the branching.
(iv) The probability for branching of a sprout tip is P_b.

Capillary sprout branching can happen if and only if these four conditions are true simultaneously.

4. Parameters estimation and computer simulation

In the simulation, we consider that, the tissue is represented by a 2-D grid. The domain space of CA is [0, 1] X [0, 1]. We assume that, the spatial domain is discretized as $x = mh$, $y = nh$ (space step h), and the temporal domain $t = k\tau$; where m, n, k, h, and τ are positive parameters. We also assume, the spatial domain length is 2 mm. and the step size *i.e.*, $\Delta x = \Delta y = h = 0.005$ and temporal step size $\tau = 0.10$. Hence, a unit of spatial step size represents 10 µm of dimensional length which is equivalent to the diameter of one or two ECs.

We assume that the pre-existing blood vessels can be represented by a straight line through the points (0, 0) and (0, 1), and the new capillary sprouts can be generated randomly from any point on the straight line; therefore, $[\mathrm{Ec}]_{i,j}^{t=0} = 1$, where$(i, j) \in [0] \times [0, 1]$. We also consider that initially MMPs are not present in the spatial domain, but the tip cell of capillary sprouts secrets MMPs, hence, at $t = 0$ MMPs is initialized with 0. We solve the system Eqs. (1)–(14) along with zero-flux boundary conditions. All the parameter values used for the simulation are dimensionless and described in Table 1. Most of the parameters are set by trial and error as it is very difficult to find out from the previous literatures. At each time step, the entire grid cells of VEGF, FGF, PDGF, ANG, EC, fibronectin, MMPs, and capillary sprout tips are updated simultaneously. The algorithm of this simulation is presented below. The model is studied with different values of w_1, w_2, and w_3 to observe the effects of chemotaxis, haptotaxis, and randomness on the sprout tip-cell movements.

Table 1
Value of the base parameters for simulating tumour angiogenesis model.

Parameter	Value	Refs.
σ_1	1.00	[1]
σ_2	0.95	
σ_3	0.90	
σ_4	0.85	
σ_5	1.00	[1]
ε_1	0.45	[1]
ε_2	0.40	
ε_3	0.35	
ε_4	0.30	
ε_5	0.45	[1]
α_1	0.001	
α_2	0.10	
α_3	0.001	
α_4	0.09	
α_5	0.001	
α_6	0.08	
α_7	0.001	
α_8	0.075	
ω_1	0.05	
ω_2	0.10	
γ_1	0.01	
γ_2	0.10	
γ_3	0.001	
λ	0.10	
φ_1	0.50	
φ_2	0.25	
φ_3	0.15	
φ_4	0.10	
P_a	0.001	
P_b	0.005	
$[\mathrm{Ec}]_{th}$	0.10	
age_{th}	1.00	

Algorithm 1
CA based tumour angiogenesis model.

Step 1: Initialize the grid size (m, n), h, τ, T, $t = 0$, and all the parameters mention in the Table 1.
Step 2: Create 2-D grids for all the chemical components, EC, and tip cells. Initialize them with the corresponding functions or values.
Step 3: Update all the grid points simultaneously for all the Chemical factors and ECs according to Eqs. (1)–(12).
Step 4: Update the sprout tip cells according to Eqs. (13)–(14).
Step 5: Create loop with i^{th} and j^{th} tip cell if, all the conditions of anastomoses are fulfilled.
Step 6: Create branches from the lth tip cell if, all the conditions of branching are fulfilled.
Step 7: $t = t + \Delta t$.
Step 8: If $t \leq T$, then go to **Step 3**. Otherwise, go to **Step 9**.
Step 9: Stop.

5. Simulating with the base parameters

To verify whether this CA based model can capture the key characteristics of capillary vessel growth, the algorithm is simulated with the base parameters mentioned in Table 1. We observe the behavior of the model in different microenvironments. Particularly, we investigate the roles of chemotaxis, haptotaxis, and randomness in the capillary vessel dynamics. Therefore, we have tested our model with four different set (w_1, w_2, w_3) of values. The description of these tests are as follows,

Experiment 1: $w_1 = 0.33$, $w_2 = 0.33$, and $w_3 = 0.33$; *i.e.*, equal weightages are given to chemotaxis, haptotacxis, and randomness.

Experiment 2: $w_1 = 0.0$, $w_2 = 0.50$, and $w_3 = 0.50$; *i.e.*, no chemotactic force is present in the domain but equal weightages are given to haptotacxis, and randomness.

Experiment 3: $w_1 = 0.50$, $w_2 = 0.0$, and $w_3 = 0.50$; *i.e.*, no haptotactic force is present in the domain but equal weightages are given to chemotaxis, and randomness.

Experiment 4: $w_1 = 0.50$, $w_2 = 0.50$, and $w_3 = 0.0$; *i.e.*, no randomness is present in the domain but equal weightages are given to chemotaxis and haptotacxis.

Our aim is to ensure that this CA model produces realistic outcome and it helps us to gain insight into the underlying biology. All the experiments are performed three times, and each time capillary vessels are created from the existing blood vessels randomly. We have also quantify the capillary vessel branching and anastomoses for these cases.

Figs. 4 and 5 show one of the simulation results (among three simulations) for the Experiment 1. The experiment begins ($t = 0$) with described VEGF, FGF, PDGF, ANG, fibronectin and EC distributions. The capillary sprouts are developed randomly from the existing blood vessels by stimulation of the surrounding TAFs (VEGF, FGF, PDGF, and ANG). As the growing capillary vessels move toward the tumour (is not shown in the Fig. 4 as the tumour is static as per our consideration), it inclined to each other and the sprout tips fuse together to create loops (anastomoses). Often these vessels split into two new branches. The number of anastomoses and branches increase as the sprout tip approaches near to the tumour (Fig. 6(a, b)). Due to the stimulation of TAFs, ECs of capillary vessels secrete MMPs which remodels the fibronectin in the ECM. The simulation is performed up to $T = 400$.

We performed Experiment 2 which produces capillary vessels but it cannot migrate toward the tumour (Fig. 7) due to the absence of chemotactic stimuli (TAFs). Hence, from this event, we can conclude that, chemotaxis is must be a necessary force for the successful vascularization in tumour microenvironment.

After the simulation of Experiment 3, we found that, it produces almost same kind of results of Experiment 1. The trajectory of the tip cell of capillary vessels are given in Fig. 8. This event indicates that without haptotaxis capillary vessels follow the same kind of trajectory as Experiment 1. The distributions of TAFs, fibronectin, MMPs, and ECs are shown in Fig. 9. In Fig. 9, we can see that, average number of loops created In Experiment 3 is almost double than Experiment 1 near to the tumour centre and average number of branches is also higher than Experiment 1.

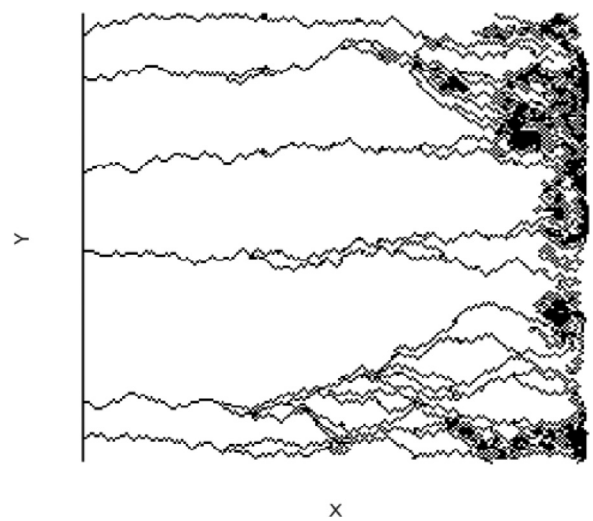

Fig. 4. Trajectory of the growing sprout tip cells of capillary vessels at $t = 400$ in Experiment 1.

We performed Experiment 4 that produces the *zig-zag* trajectory of capillary sprouts which is not realistic at all. Hence, conclusion has been drawn that, randomness is an essential feature for this model.

6. Estimation of the fractal dimensions

In this section we estimate the fractal dimensions of the growing capillary vessel networks developed by the CA model. In this research, a set of rules are applied for updating the grid points of the domain. These rules create fractal patterns by the local interactions of several components that are considered in this paper. CA provide a framework to model and explore the complexity of this dynamical system. Many researchers [58,45,50,52] have shown that, there is a direct relation between the sprouting angiogenesis and fractal patterns. Moreover, Willson [59], Culik and Dube [46], Anitas and Slyamov [60] have investigated the relationship between fractal and CA to provide a united framework for mathematical modelling of dynamic systems. The fractal dimension of a pattern illustrates the degree to which the pattern fills the space available and contrary to the notion of density which explain the mass distribution in the spatial domain, i.e., the homogeneity and compactness (higher fractal dimension) or heterogeneity and fragmentation (lower fractal dimension) of the pattern [61].

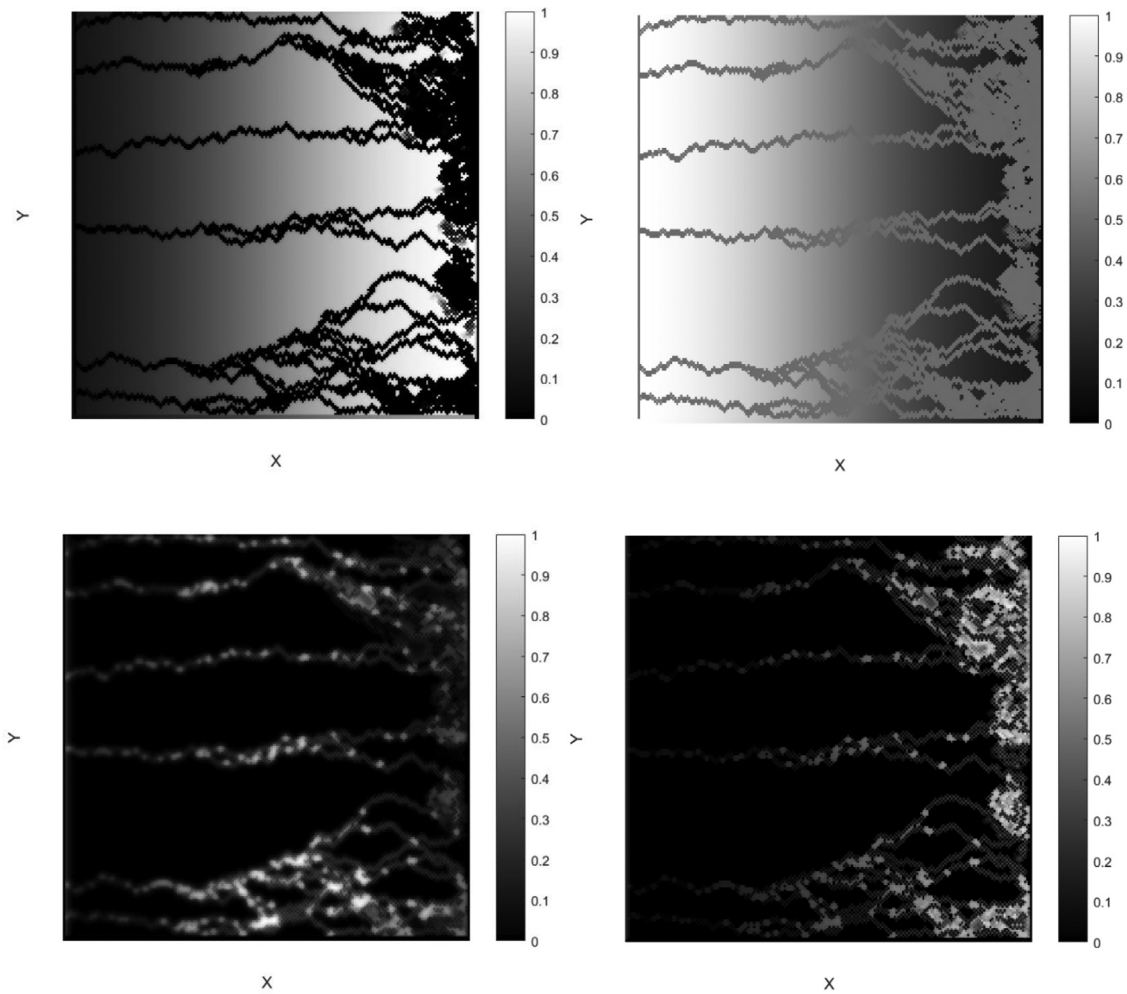

Fig. 5. Simulation results of Experiment 1: The distribution of VEGF, fibronectin (1st row), MMPs, and ECs (2nd row) are shown. All the TAFs distribution are similar for every time instances. Hence, only the distribution of VEGF is shown for different time instances.

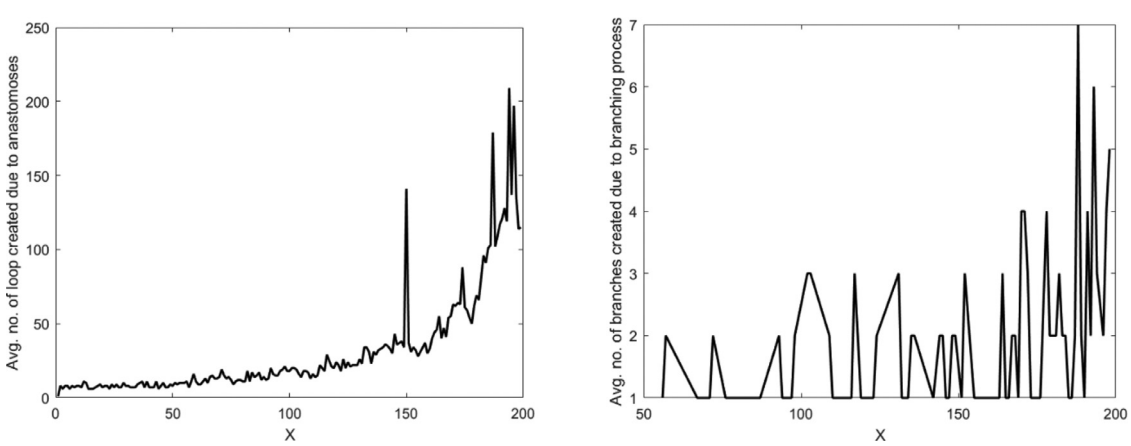

Fig. 6. Number of anastomoses (left) and branches (right) created in Experiment 1 with respect to spatial X-axis direction.

The fractal dimension can be determined using several techniques but in this study we have chosen the box-counting method. In the box-counting method, an image is covered with a grid. Then it is determine the number of boxes that require to cover the whole image. This number could be an integer or a real number. This process continues with finer grids by shrinking the grid size repeatedly. At the end of this process, the image is captured by a more accurate grid of equal size where the grid size is the dimension of the image. Let assume that a figure is covered with n number of squares of fixed size (r). Then the relation between n and r is expressed as (15),

$$n(r) = n_0 r^{-d} \tag{15}$$

In the Eq. (15) [47], $d \leq D$, where d is Minkowski-Bouligand dimension, and D is the spatial dimension (in our case, $D = 2$) and n_0 is a constant.

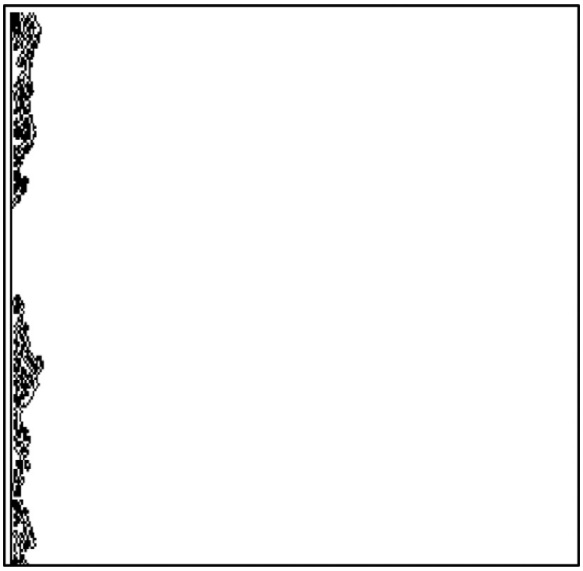

Fig. 7. Trajectory of the growing sprout tip cells of capillary vessels at $t = 400$ in Experiment 2.

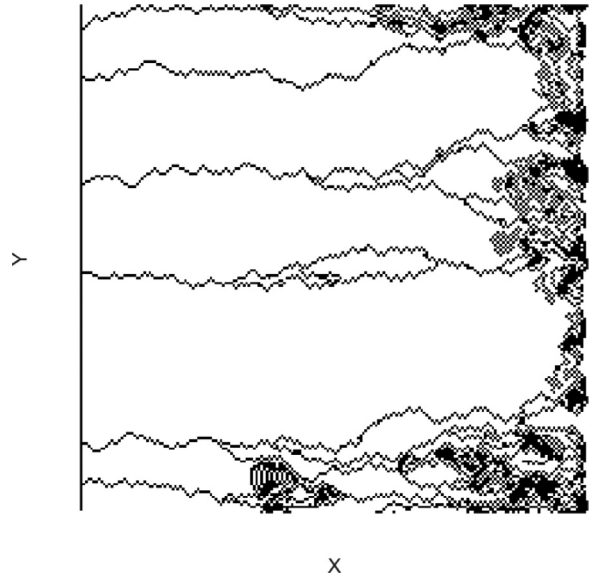

Fig. 8. Trajectory of the growing sprout tip cells of capillary vessel at $t = 400$ in Experiment 3.

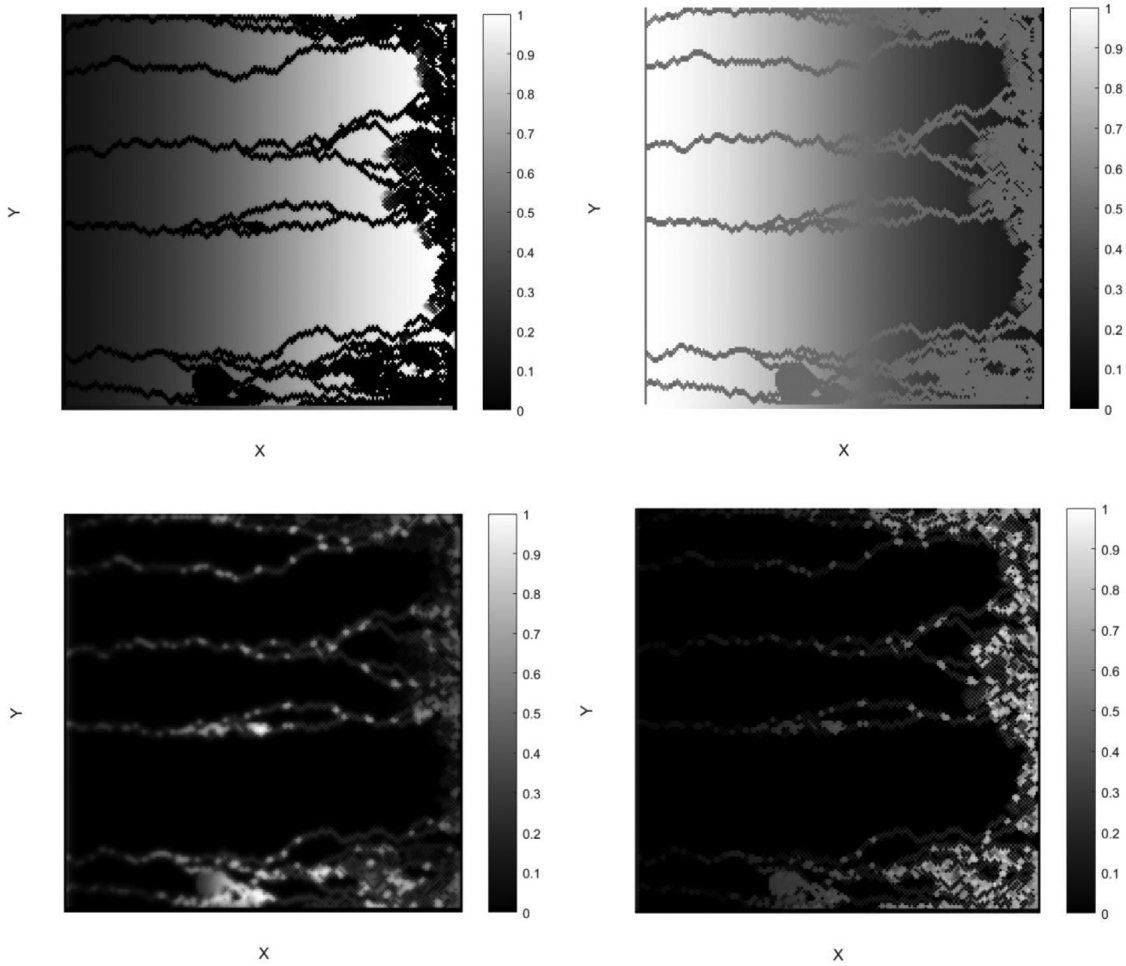

Fig. 9. Simulation results of Experiment 3: The distribution of VEGF, fibronectin (1st row), MMPs, and ECs (2nd row) are shown. All the TAFs distribution are similar for every time instances. Hence, only the distribution of VEGF is shown for different time instances.

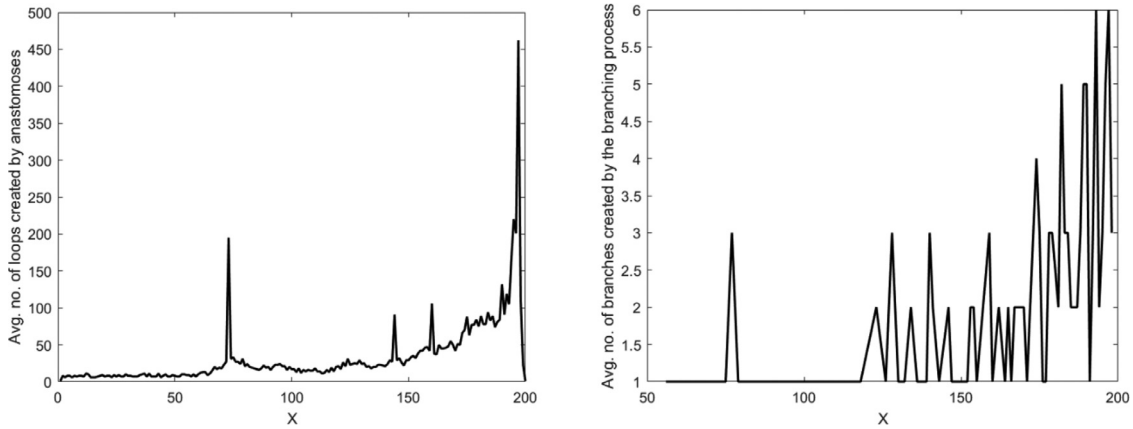

Fig. 10. Number of anastomoses (left) and branches (right) created in Experiment 3 with respect to spatial X-axis direction.

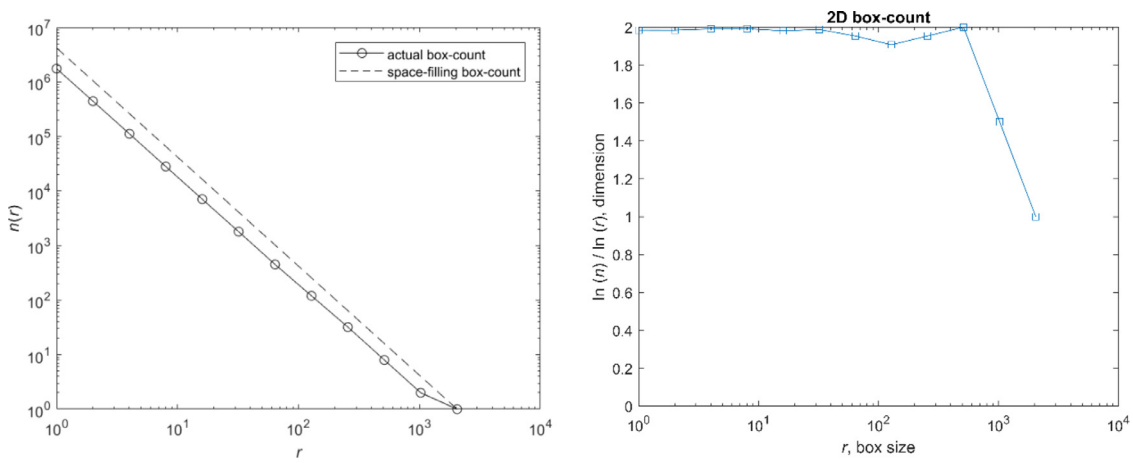

Fig. 11. The box-count (left) and dimension (right) are plotted for Experiment 1.

Moreover, fractal dimension can be determine by the slope of the line after plotting the value of$\log_e(n)$ on the Y-axis against the value of$\log_e(r)$on the X-axis. Then the dimension d of the image can be determine as,

$$d = \frac{\log_e(n)}{\log_e(r)} \tag{16}$$

Eq. (16) is adopted from (NIH [62]). For a 2-D images, the slope (or dimension of the image) lies between 1 and 2 that denotes the range between a straight line and a curve that completely fills up a 2-D plane. If the slope is steeper then it indicates the object is more fractal in nature whereas, a lower valued slope specifies that the object is closer to a straight line or less fractal nature.

In this paper, we estimate the fractal dimensions of the results produced by the Experiment 1, and the Experiment 3. The figures of the trajectories of capillary vessels (network of capillary vessels), produced by these two experiments are considered here. In Figs. 11, and 12, number of box-count with respect to box size and the local dimension with respect to box size are plotted respectively. The box-count vs. box-size plots show the discrepancy between the dotted and the solid lines in both cases that indicates a fractal behavior. The dotted line represents the scaling $n(r) = \mathrm{r}^{-2}$, for filling a 2-D image. We also plot the slope vs. box-size (r) and determine the dimension. We found from the local slope that both the figures are approximately fractal for scales $r < 100$. For the Experiment 1, we found$d = 1.9552 \pm 0.044951$ and for Experiment 3, we found $d = 1.9553 \pm 0.045014$,that are very close to the previously estimated range 1.89 ± 0.04 [45].

7. Sensitivity analysis

For validating the model, one of the standard procedure is sensitivity analysis [63], Sadhukhan and Basu [64]. Purpose of the sensitivity analysis is to quantify the robustness in the model due to the input parameters [65]. It is used to study the outcome of a model by changing one or more number of input parameters. If model's outcome varies significantly with the changes in input parameter(s), then it could be concluded that the model is strongly sensitive to the parameter(s) [66,67,68,69]. The high sensitivity demands further refinement of the model. There are various types of sensitivity analysis methods available that are different in terms of applicability, computational issues, complexity of their applications, and representations.

In this article, a partial sensitivity analysis is performed by quantifying the outcome of our model by varying some of the input parameter(s) while keeping other parameters fixed. Here, we quantify the morphological characteristics of vessels network by measuring *number of anastomoses, number of branches, number of final capillary sprouts*, and *mean fractal dimension* in the Experiment 1. The measurements have been taken by varying the values of the parameters $\alpha_1 - \alpha_8$, $\omega_1 - \omega_2$, $\gamma_1 - \gamma_3$, and λ respectively. These parameters are varied by $\pm 5\%$, $\pm 10\%$ while keeping

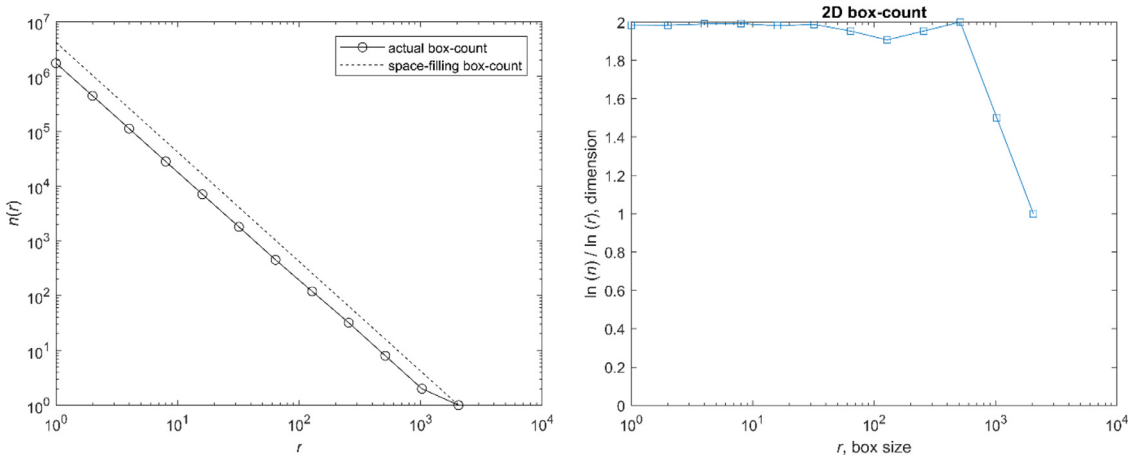

Fig. 12. The box-count (left) and dimension (right) are plotted for Experiment 3.

Table 2
Measuring RMSD for morphological characteristics of vascular network by changing $\alpha_1 - \alpha_8$, $\omega_1 - \omega_2$, $\gamma_1 - \gamma_3$, and λ.

Morphological Characteristics	+5%	+10%	Without changing parameters	−5%	−10%	RMSD	RMSD (in%)
No. of anastomoses	5261	4972	5528	5230	6064	370.96	6.85
No. of branches	130	116	125	132	133	6.64	5.22
No. of final capillary sprouts	18	20	20	18	28	3.70	17.83
Mean fractal dimension	1.9484	1.9492	1.9552	1.9487	1.9869	0.014	1.48

Table 3
Measuring RMSD for morphological characteristics of capillary sprouts by changing P_b.

Morphological Characteristics	+5%	+10%	Without changing parameters	−5%	−10%	RMSD	RMSD (in%)
No. of anastomoses	4767	7658	5528	7425	5837	1118.39	17.91
No. of branches	125	205	125	175	133	32.10	21.04
No. of final capillary sprouts	17	31	20	25	27	4.97	20.74
Mean fractal dimension	1.9874	1.9854	1.9552	1.9863	1.9869	0.012	0.633

the other parameters fixed. Sensitivity of the model is presented by the root mean square deviation (RMSD) percentage with respect to the mean outcome of the model (Table 2). The analysis is repeated for a couple of times.

In this case, the RMSD percentage for *number of anastomoses* is 6.85, for *number of branches* is 5.22, for *number of final capillary sprouts* is 17.83, and for *mean fractal dimension* is 1.83. The RMSD percentages indicate that *number of final capillary sprouts* is moderately sensitive to the specified parameters' ($\alpha_1 - \alpha_8$, $\omega_1 - \omega_2$, $\gamma_1 - \gamma_3$, and λ) values while the other features are less sensitive to these parameters specifically the *mean fractal dimension* is very less sensitive to those parameters.

Effect of P_b

P_b represents probability of branching of a mature capillary sprout. In Table 3, we have quantified the same morphological characteristics of capillary vessel network as Table 2. In this case, the other parameters are fixed and P_b is varied by ±5%, and ± 10%, respectively.

Here, the RMSD percentage for *number of anastomoses* is 17.91, for *number of branches* is 21.04, for *number of final capillary sprouts* is 20.74, and for *mean fractal dimension* is 0.6333. The RMSD percentages indicate that *number of anastomoses, number of branches*, and *number of final capillary sprouts* are moderately sensitive to the parameter (P_b) value while *mean fractal dimension* is very less sensitive.

Effect of $[Ec]_{th}$

$[Ec]_{th}$ represents the threshold concentration of ECs required for branching if the other conditions (mentioned in Section 3) are fulfilled. In Table 4, we have measured the same morphological characteristics as Tables 2 and 3 by only changing $[Ec]_{th}$. In this case, the other parameters are fixed and $[Ec]_{th}$ is varied by ±5%, and ± 10% respectively.

Here, the RMSD percentage for *number of anastomoses* is 15.97, for *number of branches* is 18.06, for *number of final capillary sprouts* is 17.60 and for *mean fractal dimension* is 0.13. The RMSD percentages specify that *number of anastomoses, number of branches, number of final capillary sprouts* are moderately sensitive to the parameter ($[Ec]_{th}$) values except *mean fractal dimension* like the two previous cases.

Effect of age_{th}

age_{th} represents the threshold age of a sprout tip cell to become mature and eligible for branching if other constraints (discussed in Section 3) are satisfied. In Table 5, we have quantified the same morphological characteristics as Tables 2, 3 and 4 by only varying age_{th} by ±5%, and ± 10% respectively.

Here, the RMSD percentage for *number of anastomoses* is 6.21, for *number of branches* is 7.18, for *number of final capillary sprouts* is 15.21 and for *mean fractal dimension* is 0.89. The RMSD percentages indicate that *number of final capillary sprouts* is moderately

Table 4
Measuring RMSD for morphological characteristics of capillary sprouts by changing $[Ec]_{th}$.

Morphological Characteristics	+5%	+10%	Without changing parameters	−5%	−10%	RMSD	RMSD (in%)
No. of anastomoses	4434	6868	5528	5528	4629	862.01	15.97
No. of branches	106	167	125	125	104	22.65	18.06
No. of final capillary sprouts	21	29	20	20	20	3.52	17.60
Mean fractal dimension	1.9489	1.9479	1.9552	1.9486	1.9486	0.0027	0.13

Table 5
Measuring RMSD for morphological characteristics of capillary sprouts by changing age_{th}.

Morphological Characteristics	+5%	+10%	Without changing parameters	−5%	−10%	RMSD	RMSD (in%)
No. of anastomoses	5528	4886	5528	5825	5825	342.96	6.21
No. of branches	125	115	125	139	139	9.24	7.18
No. of final capillary sprouts	20	18	20	26	26	3.34	15.21
Mean fractal dimension	1.9486	1.9487	1.9552	1.9866	1.9866	0.017	0.89

sensitive to the specified parameter (age_{th}) values but the other characteristics are less sensitive to this parameter.

From the Tables 2, 3, 4, and 5, we can conclude that, *mean fractal dimension* vary less compare to the other measures in all the cases discussed above. This event indicates that, morphology of capillary vessels is not much dependent on these parameters.

8. Discussion and conclusions

Angiogenesis is a critical phase in cancer as it acts as a bridge between the avascular and vascular phases. In the avascular phase, tumour growth is restricted due to limited oxygen and nutrient supply. At this moment, mainly three types of tumour cells are observed. The outer surface of the tumour is covered with proliferative cells as it absorbs most of oxygen and nutrients. While the inner area of tumour is mostly filled up with necrotic cells due to deprivation of nutrients and oxygen. The quiescent cells are constituted in between proliferative and necrotic cells. As the oxygen and nutrient deficiency increases within the tumour microenvironment, quiescent and proliferative cells gradually turn into hypoxic and then convert into necrotic cells. To overcome this situation, hypoxic cells secrete several TAFs which diffuse into the microenvironment and attract the nearby blood vessels. During tumour culture, TAFs are mainly found in the border area of necrotic and quiescent layers. Therefore, it can be said that, in tumour angiogenesis, different tumour cells and their dynamics have important roles in the production of TAFs. Moreover, tumour cell dynamics are directly dependent on oxygen and nutrient supply.

It was hypothesized that if somehow angiogenesis would get inhibited then the aggressiveness and the progression of a tumour would be reduced. This was the motivation for active research on tumor angiogenesis. For many years, scientists have tried to find out anti-angiogenic agents which could be useful in cancer therapy [70,71]. Capillary vessels are an obvious target for an anti-angiogenic agent. The idea behind the treatment procedure is that these agents could destroy the existing tumour blood vessels and hinder the formation of new vessels. The agents also neutralize the active proliferative endothelial cells [72] of the surrounding blood vessels. These events restrict nutrient and oxygen supply within tumour cells which makes the tumour cells necrosis.

Tumour angiogenesis phase can be divided into three subphases which occurred one after another. These phases are following:

Capillary sprout generation: Capillary sprouts are formed from the pre-existing blood vessels due to the effects of TAFs. Chemotaxis and haptotaxis help these sprouts to move towards the tumour. During its journey, it creates a complex vascular network through branching and anastomoses.

Vessels adaptation and network remodelling/pruning: The pro-angiogenic factors become more powerful during angiogenesis. As a result, newly developed vessels grow unboundedly and the vascular network becomes unorganized, often leaky, chaotic in nature, and hemorrhagic due to upregulation of growth factors. Blood flow within these networks is often slow or oscillating which makes it dysfunctional [24].

These capillary vessels experienced extensive vascular remodelling to make it functional and mature. The remodelling includes vascular pruning and regression of branches. Vessels pruning refers to the trimming of a vascular network. In this phase, a subset of growing vessels undergoes physiological regression.

Blood flow through the vascular network: Blood is a complex biphasic medium. It contains red blood cells, white blood cells, and platelets. The solid elements constitute approximately 45% of the total blood composition. Due to its biphasic properties, blood does not behave as a continuum. Vessels adaptation and network remodelling strongly affect blood flow [73,74,75,76,77]. The rheological characteristics of blood and vessel network remodelling are interlinked as blood flow creates stress on the vessels' wall that helps to adjust the vessels' diameter either due to vasodilatation or constriction. On the other hand, blood rheology is also influenced by changes in network architecture.

In this article, we focus only on the initial phase of angiogenesis, *i.e.*, capillary sprouts formation from the pre-existing blood vessels and its movement towards the tumour which creates a complex vascular network. The other phases of tumour angiogenesis are not in the scope of this manuscript but we keep them as our future work. Unfortunately, very less information is available regarding the processes involved in tumour angiogenesis. In this article, a CA-based discrete model is developed to make useful predictions. Using the interactions in 2-D tissue space, we study the effects of various biological factors like VEGF, FGF, PDGF, ANG, MMPs, fibronectin, and ECs during the formation of the capillary sprouts. To reduce the complexity of the model, it is considered that the tumour is a static entity (as its growth will be stagnant after it becomes 1–2 mm of diameter). The model also captures anastomoses and branching phenomena in capillary blood vessels. In each run of the simulation, the number of sprout formations and locations are varied randomly. It is observed that chemotaxis and randomness are the essential components of the capillary sprout motility whereas, haptotaxis works as an add-on feature.

As the simulation parameters may vary due to several biochemical and biophysical processes, the variability of the model is observed by changing the values of the important parameter(s). It is found that the *mean fractal dimension* varies less compare to the other measures like *number of anastomoses*, *number of branches*, *number of final capillary sprouts*. Moreover, the relation between the CA-based model and the fractal dimension is discussed. Also, fractal dimensions for the simulated outcomes are determined which are very close to the previously estimated range.

Our proposed CA-based model is not a product of any real research but an approximation of tumour angiogenesis process. Although, it could be useful for determining the efficacy of anti-angiogenic therapies. Preclinical theoretical studies of the anti-angiogenic agents explore whether the agent is properly working against tumours with the appropriate target and/or whether it could behave in the same way in the human body. To study the behaviour of these anti-angiogenic agents our model would be useful. Overall, the model justifies the practical scenario though it needs to be upgraded further with more details by including tumour cell dynamics, different phenotypes of the tumour cells, dynamics of oxygen and nutrient concentrations, capillary vessels pruning, and blood flow through the newly formed vessels, etc.

Declaration of Competing Interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

CRediT authorship contribution statement

Sounak Sadhukhan: Conceptualization, Visualization, Formal analysis, Writing – original draft, Writing – review & editing. **P.K. Mishra:** Conceptualization, Visualization, Software, Methodology, Writing – original draft, Writing – review & editing.

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Appendix

In 3-D version of the model, we consider that the tissue space is represented by a 3-D grid of cells. The domain space of CA is [0, 1] X [0, 1] X [0, 1]. Like the 2-D model, we assume that the spatial domain is discretized as $x = mh$, $y = nh$, and $z = uh$ (space step h), where $m, n, u,$ are h are the positive parameters. We consider that the spatial domain length is 2 mm. and the step size $h = 0.01$. Hence, a unit of step size represents 20 µm of dimensional length which is equivalent to the diameter of four ECs.

We assume that, the tumor is spherical in shape. The centre of the tumour is located at (0.5, 0, 0.5) in the domain and radius of the tumour is 1 mm or 50h The pre-existing blood vessels are represented by the horizontal straight lines: (0, 0, 0.2) to (1, 0, 0.2, 0, 0, 0.4) to (1, 0, 0.4, 0, 0, 0.6) to (1, 0, 0.6, 0, 0, 0.8) to (1, 0, 0.8); and the vertical straight lines: (0.3, 0, 0) to (0.3, 0, 1) and (0.6, 0, 0) to (0.6, 0, 1), respectively. Hence, each and every chemical components are initialized properly using the Eqs. (1a)–(5a).

$$[\mathrm{C_V}]^{t=0}_{x,y,z} = \sigma_1 \exp\left(\frac{-(1-y)^2}{\varepsilon_1}\right), (x,y,z) \in [0,1]\times[0,1]\times[0,1] \tag{1a}$$

$$[\mathrm{C_F}]^{t=0}_{x,y,z} = \sigma_2 \exp\left(\frac{-(1-y)^2}{\varepsilon_2}\right), (x,y,z) \in [0,1]\times[0,1]\times[0,1] \tag{2a}$$

$$[\mathrm{C_P}]^{t=0}_{x,y,z} = \sigma_3 \exp\left(\frac{-(1-y)^2}{\varepsilon_3}\right), (x,y,z) \in [0,1]\times[0,1]\times[0,1] \tag{3a}$$

$$[\mathrm{C_A}]^{t=0}_{x,y,z} = \sigma_4 \exp\left(\frac{-(1-y)^2}{\varepsilon_4}\right), (x,y,z) \in [0,1]\times[0,1]\times[0,1] \tag{4a}$$

and

$$[\mathrm{C_f}]^{t=0}_{x,y,z} = \sigma_5 \exp\left(\frac{-y^2}{\varepsilon_5}\right), (x,y,z) \in [0,1]\times[0,1]\times[0,1] \tag{5a}$$

The new capillary sprouts can be generated randomly from any point on the existing blood vessels. We consider that initially MMPs are not present in the spatial domain, but the tip cell of capillary sprouts secrets MMPs, hence, at $t = 0$ MMPs is initialized with 0 for all $(x, y, z) \in$ [0,1]x[0,1]x[0,1].

We use Moore's neighborhoods for determining the value of the species at each location. For example, Mo($A_{i,j,k}$) refers to the Moore's neighborhood elements of $(i, j, k)^{\text{th}}$ position in an arbitrary 3-D tissue space A, i.e., $A_{i+1,j,k}$, $A_{i-1,j,k}$, $A_{i,j+1,k}$, $A_{i,j-1,k}$, $A_{i,j,k+1}$, $A_{i,j,k-1}$, $A_{i+1,j+1,k}$, $A_{i+1,j-1,k}$, $A_{i+1,j,k+1}$, $A_{i+1,j,k-1}$, $A_{i-1,j+1,k}$, $A_{i-1,j-1,k}$, $A_{i-1,j,k+1}$, $A_{i-1,j,k-1}$, $A_{i+1,j+1,k+1}$, $A_{i+1,j+1,k-1}$, $A_{i+1,j-1,k+1}$, $A_{i+1,j-1,k-1}$, $A_{i-1,j+1,k+1}$, $A_{i-1,j+1,k-1}$, $A_{i-1,j-1,k-1}$, $A_{i-1,j-1,k+1}$, $A_{i,j+1,k+1}$, $A_{i,j+1,k-1}$, $A_{i,j-1,k+1}$, $A_{i,j-1,k-1}$. From the above considerations, the following CA rules (6a – 11a) are formulated.

$$[\mathrm{C_V}]^{t+1}_{i,j,k} = [\mathrm{C_V}]^{t}_{i,j,k} + \alpha_1 \mathrm{avg}\left(\mathrm{Mo}\left([\mathrm{C_V}]^{t}_{i,j,k}\right)\right) - \alpha_2 [\mathrm{C_V}]^{t}_{i,j,k}[\mathrm{Ec}]^{t}_{i,j,k} \tag{6a}$$

$$[\mathrm{C_F}]^{t+1}_{i,j,k} = [\mathrm{C_F}]^{t}_{i,j,k} + \alpha_3 \mathrm{avg}\left(\mathrm{Mo}\left([\mathrm{C_F}]^{t}_{i,j,k}\right)\right) - \alpha_4 [\mathrm{C_F}]^{t}_{i,j,k}[\mathrm{Ec}]^{t}_{i,j,k} \tag{7a}$$

$$[\mathrm{C_P}]^{t+1}_{i,j,k} = [\mathrm{C_P}]^{t}_{i,j,k} + \alpha_5 \mathrm{avg}\left(\mathrm{Mo}\left([\mathrm{C_P}]^{t}_{i,j,k}\right)\right) - \alpha_6 [\mathrm{C_P}]^{t}_{i,j,k}[\mathrm{Ec}]^{t}_{i,j,k} \tag{8a}$$

$$[\mathrm{C_A}]^{t+1}_{i,j,k} = [\mathrm{C_A}]^{t}_{i,j,k} + \alpha_7 \mathrm{avg}\left(\mathrm{Mo}\left([\mathrm{C_A}]^{t}_{i,j,k}\right)\right) - \alpha_8 [\mathrm{C_A}]^{t}_{i,j,k}[\mathrm{Ec}]^{t}_{i,j,k} \tag{9a}$$

$$[\mathrm{C_f}]^{t+1}_{i,j,k} = [\mathrm{C_f}]^{t}_{i,j,k} + \omega_1 [\mathrm{Ec}]^{t}_{i,j,k} - \omega_2 [\mathrm{C_f}]^{t}_{i,j,k}[\mathrm{Ec}]^{t}_{i,j,k} \tag{10a}$$

$$\begin{aligned}[\mathrm{C_M}]^{t+1}_{i,j,k} &= [\mathrm{C_M}]^{t}_{i,j,k} + \gamma_1 \mathrm{avg}\left(\mathrm{Mo}\left([\mathrm{C_M}]^{t}_{i,j,k}\right)\right) \\ &\quad + \gamma_2 \psi(i,j,k)[\mathrm{Ec}]^{t}_{i,j,k} - \gamma_3 [\mathrm{C_M}]^{t}_{i,j,k}[\mathrm{Ec}]^{t}_{i,j,k}\end{aligned} \tag{11a}$$

In (6a)–(11a), α_1, α_2, α_3, α_4, α_5, α_6, α_7, α_8, ω_1, ω_2, γ_1, γ_2, and γ_3 are the positive parameters like the 2-D model. In Eq. (11a), $\psi(.)$ is a function that returns 1, if grid position (i, j, k) contains an endothelial cell otherwise it returns 0.

Moreover, it is considered that EC concentration is updated by the corresponding six orthogonal grid points only. Hence, for updating EC concentration, we are taking the average of its von Neumann neighborhood. Therefore, in (12a), vN($A_{i,j,k}$) refers to the von Neumann's neighbors of $(i, j, k)^{\text{th}}$ position in the 3-D grid A, i.e., $A_{i+1,j,k}$, $A_{i-1,j,k}$, $A_{i,j+1,k}$, $A_{i,j-1,k}$, $A_{i,j,k+1}$, and $A_{i,j,k-1}$. Therefore, ECs are updated as,

$$\begin{aligned}[\mathrm{Ec}]^{t+1}_{i,j,k} &= [\mathrm{Ec}]^{t}_{i,j,k} + \lambda \mathrm{avg}\left(\mathrm{vN}\left([\mathrm{Ec}]^{t}_{i,j,k}\right)\right) \\ &\quad .\left(\varphi_1 [\mathrm{C_V}]^{t}_{i,j,k} + \varphi_2 [\mathrm{C_F}]^{t}_{i,j,k} + \varphi_3 [\mathrm{C_P}]^{t}_{i,j,k} + \varphi_4 [\mathrm{C_A}]^{t}_{i,j,k}\right)\end{aligned} \tag{12a}$$

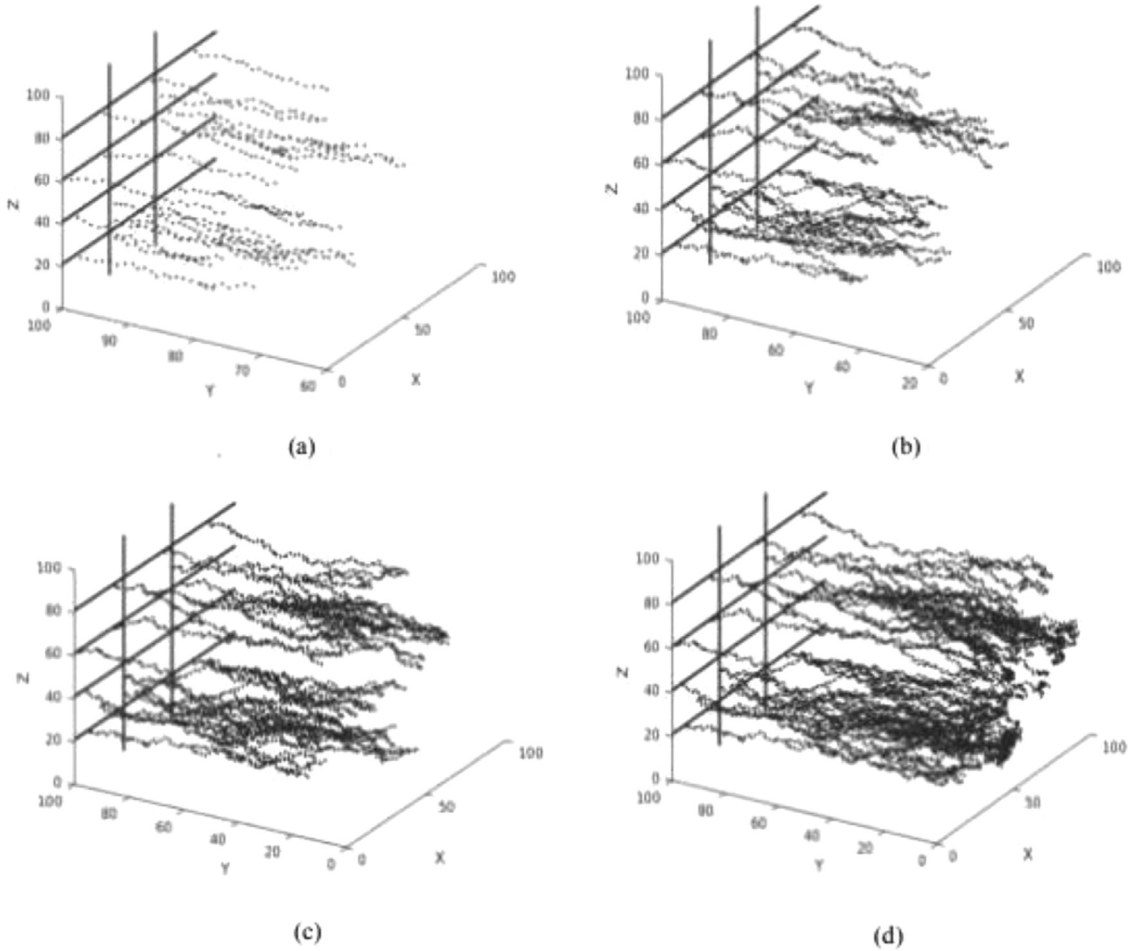

Fig. A. Capillary sprouts formation in tumour angiogenesis due to chemotaxis, haptotaxis, and random motility on (a) 5th, (b) 10th, (c) 15th, and (d) 20th days respectively in 3-D domain space. The model is simulated with $w_1 = 0.33$, $w_2 = 0.33$, and $w_3 = 0.33$.

In (12a), λ, φ_1, φ_2, φ_3, and φ_4 are the positive parameters. φ_1, φ_2, φ_3, and φ_4 are the weightage given to the respective TAFs considered here, such that, $\varphi_1 > \varphi_2 > \varphi_3 > \varphi_4$, and $\varphi_1 + \varphi_2 + \varphi_3 + \varphi_4 = 1$.

The movement of these capillary sprouts are controlled by their corresponding tip cells. We consider that every iteration is equivalent to 2.4 h. and a sprout tip cell requires 10 iterations ($\approx$ 24 h = 1 day) to become mature. The newly developed capillary sprouts (S) move towards the tumour due to chemotactic force (for the higher concentration of TAFs) while fibronectin increases the adhesiveness between the sprouts and the ECM that generate a haptotactic force between them. Therefore, the movement of capillary sprouts are controlled by the chemotactic as well as the haptotactic forces. As the capillary sprouts approach toward the tumour, it can move either any of the adjacent twenty six direction (Moore neighbors) or it may remain idle at the same position. In this context, randomness is also included by incorporating a random variable. Considering all the mentioned features above, we produce Eqs. (13a) and (14a) as,

Where,

$$g^t_{i,j,k} = \underbrace{w_1\left(\varphi_1[C_V]^t_{i,j,k} + \varphi_2[C_F]^t_{i,j,k} + \varphi_3[C_P]^t_{i,j,k} + \varphi_4[C_A]^t_{i,j,k}\right)}_{\text{chemotaxis}} + \underbrace{w_2[Ec]^t_{i,j,k}[C_f]^t_{i,j,k}}_{\text{haptotaxis}} + \underbrace{w_3 pd_{i,j,k}}_{\text{randomness}} \tag{14a}$$

Eq. (13a) indicates that, the sprout tip located at (i, j, k) will move at next time instance to the location that has maximum 'g' value among the Moore's neighbors. If the Eq. (13a) returns (i, j, k), then it indicates that the sprout tip will be in idle position otherwise the sprout tip moves to a neighboring specified position. In (14a), w_1, w_2, and w_3 are the corresponding weightage given to chemotaxis, haptotaxis, and randomness respectively, where, $w_1 + w_2 + w_3 = 1$. Here, p is a random variable, which infuses randomness in the equation, where,$p \in [0, 1]$ with uniform

$$\text{Direction of}\left(S^{n+1}_{i,j,k}\right) = \text{index of}\left\{\max\begin{pmatrix} g^t_{i,j,k}, g^t_{i+1,j,k}, g^t_{i-1,j,k}, g^t_{i,j+1,k}, g^t_{i,j-1,k}, g^t_{i,j,k+1}, g^t_{i,j,k-1}, \\ g^t_{i+1,j+1,k}, g^t_{i+1,j-1,k}, g^t_{i-1,j+1,k}, g^t_{i-1,j-1,k}, g^t_{i+1,j,k+1}, g^t_{i+1,j,k-1}, \\ g^t_{i-1,j,k+1}, g^t_{i-1,j,k-1}, g^t_{i,j+1,k+1}, g^t_{i,j+1,k-1}, g^t_{i,j-1,k+1}, g^t_{i,j-1,k-1}, \\ g^t_{i+1,j+1,k-1}, g^t_{i+1,j+1,k+1}, g^t_{i+1,j-1,k+1}, g^t_{i+1,j-1,k-1}, g^t_{i-1,j+1,k+1}, \\ g^t_{i-1,j+1,k-1}, g^t_{i-1,j-1,k+1}, g^t_{i-1,j-1,k-1} \end{pmatrix}\right\} \tag{13a}$$

distribution, and $d_{i,j,k}$ is the reciprocal of Euclidian distance between the tumour centre (0.5, 0, 0.5) and the point (i, j, k). The rules for branching and anastomoses is same as the 2-D model. The presented 3-D model is simulated with the values described in Table 1 (in the manuscript) and $w_1 = 0.33$, $w_2 = 0.33$, and $w_3 = 0.33$. The outcomes of the simulated model are visualized in the following Fig. A (a–d).

All the grid elements are updated concurrently. At each time step, we update the values of VEGF, FGF, PDGF, ANG concentrations, ECs density, fibronectin, MMPs, and the location of capillary sprout tips. The 3-D model is simulated to observe the effects of chemotaxis, haptotaxis, and random motility on the sprout tip-cell movement and morphology of the capillary vessels on 5th, 10th, 15th, and 20th days (Fig. (a)–(d)), respectively. Through from the simulation, a number of observations are made: (a) no two capillary sprouts are generated from adjacent locations; (b) as the sprouts approach closer to the tumour, its branching tendency increases.

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